

Medication adherence is a critical factor in managing chronic illnesses, yet many patients forget to take their prescribed doses on time. This can lead to serious health complications, reduced treatment effectiveness and increased medical costs. Traditional pillboxes rely on manual tracking, which can be ineffective with memory challenges or busy schedules.

This project aims to develop a smart pillbox using an Arduino and an MC-38 magnetic sensor to monitor whether the pillbox has been opened within a specified timeframe. If the pillbox remains unopened past the scheduled medication time, the system will trigger an alert (such as a buzzer, LED, or mobile notification) to remind the user to take their medication. By automating adherence tracking, this smart pillbox can help patients manage their medication schedules more effectively, improving health outcomes and reducing the risk of missed doses.

1. For our requirements, we needed the pill box to fit the user's medication, to notify users if they haven't taken their medication within 24 hours, and to be able to fit onto the box to optimize space. For our box to notify the user, we need to create and test the code for our screen that's on the front of the Arduino to show that either A: the person has taken their medication or B: if they need to take it. We would use an integer representation in the code and have it set on a specific analogue clock. When the timer reaches a certain time, for example, 2:30 pm, the box's screen would change to show that the user needs to take their medication. Once the box is open, we would use a door sensor for the clock to reset back to 0 so it can count back up to 24 hours and show that the user has opened their pill box within that timeframe. For the second requirement, we need the motherboard and LCD for the Arduino to fit into the box to optimize space. We created a holder on the front of it for the wires, motherboard, and a hole where the ground wire and hinge sensors can connect to the device and power.
2. For our testing procedure, we made sure the code would work with our final design and corrections that we needed to make, our highest priority was that the box had to notify the user in a timely manner that was easy to see so we added the LCD onto the design, our second highest was that the box had to be big enough to fit the users medications, and our least important priority was that the mechanics must fit into the box to optimize space.

3.

Data

Attempt 1	Box sensed when it opened/closed	Nothing appeared on the screen	Making adjustments to the screen
Attempt 2		The screen showed binary code	May have gotten wires confused
Attempt 3	Box sensed	The screen was	The code was in a

	opened/closed	flashing	loop
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4. For our conclusion, the screen was broken in every LDC we tried. Our solution was a good idea, but the screen would need to be constantly updated just to make sure it worked and it wouldn't break. The next step to improve this would be to add a screen that lasts longer and is cheaper to produce.